

Tutorial 2

Check your knowledge

- d** 1. An airspeed indicator shows the _____ based on pressure measurements taken from a _____.
- (a) “ground speed” - “static port”
 - (b) “equivalent airspeed” - “static port”
 - (c) “true airspeed” - “pitot tube”
 - (d) “equivalent airspeed” - “pitot-static tube”
- a** 2. The altimeter shows _____ based on pressure measurements taken from a _____.
- (a) “geopotential height” - “static port”
 - (b) “geometric height” - “pitot tube”
 - (c) “geometric height” - “static port”
 - (d) “density altitude” - “pitot tube”
3. Are the following statements *true* or *false*?
- Unless stated otherwise, you may assume a standard day and that gravitational acceleration is a function of altitude alone.
- T** (a) At sea level, the geometric and geopotential heights are both zero.
 - F** (b) At any given pressure level, above sea level, the geopotential height will be higher ^{less} than the geometric height. $\int_0^{ht} g \, dz = g \cdot h_g$
 - F** (c) The Standard Atmosphere model given as Table 1, indicates that at a geopotential ^{geometric} height of 20 km, the atmospheric pressure will be 5,475 Pa.
 - T** (d) The reference sea-level temperature T_0 for the ISA model is 15°C.
 - F** (e) Equivalent Airspeed readings taken by a pilot on a non-standard day ($T \neq T_0$ or $P_0 \neq 101325$ Pa) will not be correct.
 - T** (f) Using ISA tables to find the True Airspeed of an aircraft from Equivalent Airspeed readings taken on a non-standard day will yield incorrect results.
 - F** (g) The ground speed of an aircraft can be calculated by adding a known tailwind component to its equivalent ^{true} airspeed.
 - F** (h) Thrust specific fuel consumption is defined as the mass of fuel consumed per unit thrust ^{per unit time} produced.
 - F** (i) The fuel consumption of a turboshaft engine is typically expressed in terms of thrust specific ^{power} fuel consumption.
 - F** (j) The thrust produced at a given altitude is a function of the local pressure ^{only}.
 - T** (k) All other things being equal, an increase in temperature would result in a reduction of lift being generated. $P = \rho RT$
 - F** (l) Wave drag is encountered only in the supersonic ^{transonic} speed regime and arises due to the formation of shock waves over the aircraft.
 - T** (m) Wind axes are a type of body-fixed axis system, where the x -axis is aligned to the freestream flow.

Application Exercise

The Perlan II, seen in figure 1, is a pressurized sailplane designed to fly up to an altitude of 90,000 ft to conduct atmospheric research. The drag curve of the aircraft can be assumed to be parabolic and is known to be approximately

$$C_D = 0.012 + 0.013C_L^2.$$

The aircraft has a fully loaded mass of 816 kg, a wing area of 24.4 m² and a maximum lift coefficient $C_{L_{max}} = 1.4$.



Figure 1: Rendering of Perlan II in high altitude flight (Airbus)

1. Derive the 2D axial equations of motion for a glider in wings level flight. Hence show that for a steady, constant slope glide

$$\tan \gamma = -\frac{C_D}{C_L}$$

For the purposes of this question you may assume small angles.

[30%]

2. What is the minimum possible flight speed for the aircraft at 90,000 ft? Express your answer in EAS, TAS and Mach number. How would these quantities change with altitude?

[20%]

3. Starting from an altitude of 50,000 ft and maintaining a constant EAS of 54 kts, what distance over ground will the aircraft be able to cover?

[20%]

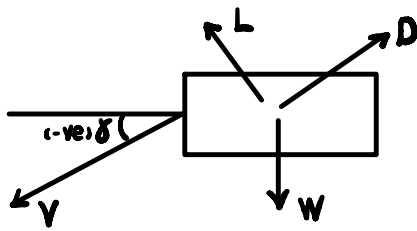
4. What lift coefficient should the pilot be operating the aircraft at to maximise the glide distance possible?

[30%]

Extra Credit

5. How might your result above differ from the lift coefficient required to achieve the maximum flight time? Is this flight condition achievable?

1.



$$L = W \cos(-\delta) = W \cos \delta = \frac{1}{2} \rho V^2 S C_L$$

$$D = W \sin(-\delta) = -W \sin \delta = \frac{1}{2} \rho V^2 S C_D$$

$$\therefore \tan \delta = -\frac{D}{L} = -\frac{C_D}{C_L}$$

$$2. \quad W = L = \frac{1}{2} \rho_0 V_s^2 S C_{L_{max}} = mg = 816 \text{ kg} \times 9.81 \text{ m/s}^2 = 8004.96 \text{ N}$$

$$\therefore V_{s, EAS} = \left(\frac{2W}{\rho_0 S C_{L_{max}}} \right)^{\frac{1}{2}} = 19.560 \text{ m/s}$$

$$\therefore \frac{1}{2} \rho V_{s, TAS}^2 = \frac{1}{2} \rho_0 V_{s, EAS}^2$$

$$\text{At } 90000 \text{ ft, } h = 27432.33 \text{ m}$$

$$\therefore V_{s, TAS} = \left(\frac{\rho_0 V_{s, EAS}^2}{\rho} \right)^{\frac{1}{2}} = \left(\frac{V_{s, EAS}^2}{6} \right)^{\frac{1}{2}} = 138.033 \text{ m/s}$$

$$\therefore M = \frac{V_{s, TAS}}{C} = \frac{138.033 \text{ m/s}}{300.5 \text{ m/s}} = 0.459$$

$$3. \quad \text{When } V_{EAS} = 54 \text{ kts} = 27.78 \text{ m/s}, h = 50000 \text{ ft} = 15240.19 \text{ m}$$

$$\tan \delta = -\frac{C_D}{C_L} = -\frac{0.012 + 0.013 C_L^2}{C_L}$$

$$\text{For } C_L, C_L = \frac{W}{\frac{1}{2} \rho_0 V_{EAS}^2 S} = 0.694$$

$$\therefore \tan \delta = -8.268 \times 10^{-3}$$

$$\therefore \tan \delta = \frac{\Delta h}{\Delta d}$$

$$\therefore d = \frac{-h}{\tan \delta} = 579195 \text{ m}$$

$$4. \therefore d = \frac{-h}{\tan \gamma}$$

$\therefore$ to maximise d , minimise $-\tan \gamma$

$$\therefore -\tan \gamma = \frac{0.012 + 0.013 C_L^2}{C_L} = \frac{0.012}{C_L} + 0.013 C_L$$

$$\therefore \frac{d(-\tan \gamma)}{dC_L} = -0.012 C_L^{-2} + 0.013 = 0$$

$$\therefore C_L = \sqrt[2]{\frac{0.012}{0.013}} = 0.961$$

5. For maximum flight time, V_i constant

$$\begin{aligned} t &= \frac{d}{V_{\text{true}} \cos \gamma} = \frac{-h \tan \gamma}{\left(\frac{2W}{\rho \cdot C_L S} \right)^{\frac{1}{2}} \cos \gamma} = \frac{h (\rho \cdot C_L S)^{\frac{1}{2}} (0.012 + 0.013 C_L^2)}{(2W)^{\frac{1}{2}} C_L \cos \gamma} \\ &= \frac{h (\rho \cdot C_L S)^{\frac{1}{2}} (0.012 + 0.013 C_L^2)}{(2W)^{\frac{1}{2}} C_L} \cdot \frac{2W}{\rho \cdot V_i^2 S C_L} = \frac{h (2W)^{\frac{1}{2}} (0.012 + 0.013 C_L^2)}{(\rho \cdot S)^{\frac{1}{2}} V_i^2 C_L^{\frac{3}{2}}} \end{aligned}$$

$$\therefore t \propto \frac{0.012 + 0.023 C_L^2}{C_L^{\frac{3}{2}}} \quad \therefore \text{let } t_{CL} = 0.012 C_L^{-\frac{3}{2}} + 0.023 C_L^{\frac{1}{2}}$$

$$\therefore \frac{dt_{CL}}{dC_L} = -\frac{9}{500} C_L^{-\frac{5}{2}} + \frac{23}{2000} C_L^{-\frac{1}{2}} = 0 \quad \therefore C_L = \frac{6}{23} \sqrt{23}$$

$$\therefore \frac{d^2 t_{CL}}{dC_L^2} = \frac{9}{200} C_L^{-\frac{7}{2}} - \frac{23}{4000} C_L^{-\frac{3}{2}} \quad \text{when } C_L = \frac{6}{23} \sqrt{23} \quad (C_L > 0)$$

$$= 0.016 > 0 \quad \therefore \text{minimum point exists}$$

$\therefore$ The flight condition is not achievable

Table 1: Atmospheric properties in the lower atmosphere (1976 COESA model)

Altitude h (m)	Temperature T (K)	Pressure P (Pa)	Density ratio σ	Speed of Sound a_s (m/s)
0	288.1	101325	1.00000	340.3
500	284.9	95461	0.95287	338.4
1000	281.6	89875	0.90746	336.4
1500	278.4	84556	0.86373	334.5
2000	275.1	79495	0.82162	332.5
2500	271.9	74683	0.78111	330.6
3000	268.6	70109	0.74214	328.6
3500	265.4	65764	0.70468	326.6
4000	262.1	61640	0.66868	324.6
4500	258.9	57728	0.63410	322.6
5000	255.6	54020	0.60091	320.5
5500	252.4	50507	0.56907	318.5
6000	249.1	47181	0.53853	316.4
6500	245.9	44035	0.50926	314.4
7000	242.6	41061	0.48123	312.3
7500	239.4	38251	0.45439	310.2
8000	236.1	35600	0.42871	308.1
8500	232.9	33099	0.40415	305.9
9000	229.6	30742	0.38069	303.8
9500	226.4	28524	0.35829	301.6
10000	223.1	26436	0.33690	299.5
10500	219.9	24474	0.31651	297.3
11000	216.6	22632	0.29708	295.1
11500	216.6	20916	0.27455	295.1
12000	216.6	19330	0.25374	295.1
12500	216.6	17865	0.23450	295.1
13000	216.6	16510	0.21672	295.1
13500	216.6	15259	0.20029	295.1
14000	216.6	14102	0.18510	295.1
14500	216.6	13033	0.17107	295.1
15000	216.6	12045	0.15810	295.1
15500	216.6	11131	0.14611	295.1
16000	216.6	10287	0.13504	295.1
16500	216.6	9508	0.12480	295.1
17000	216.6	8787	0.11534	295.1
17500	216.6	8121	0.10659	295.1
18000	216.6	7505	0.09851	295.1
18500	216.6	6936	0.09104	295.1
19000	216.6	6410	0.08414	295.1
19500	216.6	5924	0.07776	295.1
20000	216.6	5475	0.07187	295.1
22000	218.6	4000	0.05202	296.4
24000	220.6	2930	0.03777	297.8
26000	222.6	2153	0.02750	299.1
28000	224.6	1586	0.02008	300.5
30000	226.6	1172	0.01470	301.8